

Vibrational Deduction Transformer (VDT)

A Variational Autoencoder with Graph Wiring
Latent Space and Associative Spectral Memory

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Abstract

We introduce the *Vibrational Deduction Transformer* (VDT), a generative architecture that treats the learning process as a vibrational system grounded in Rayleigh’s *Theory of Sound*. The VDT is a deductive transformer that leverages feature-space graph Laplacian: the learning process is guided by the Laplacian eigenbasis as harmonics of a spring/string vibrating system endowed with a characteristic density matrix. The decoder maps a stochastic latent code $\mathbf{z}$ to a *spectral loading matrix* $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$, where $\mathbf{U}_{1:q}$ are the q lowest-frequency eigenvectors of a frozen ArrowSpace graph Laplacian index $\mathbf{L}(\mathcal{I})$, and $\mathbf{S}$ is a learnable coefficient matrix in that eigenbasis. An associative Hopfield-type memory is appended to the spectral artefact emitted after training, wiring the vibrational representation into a downstream transformer. The objective is a four-term Evidence Lower Bound (ELBO) comprising a reconstruction term, an isotropic latent KL, an eigenvalue-weighted spectral-basis KL (spectral Occam’s razor), and a Gamma-vs-Exponential mode-frequency KL for spectral band selection. Benchmarks on CORA, PubMed, and synthetic spring-network graphs demonstrate entropy-controlled spectral graph generation and strong linear-probe accuracy of frozen latent representations.

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1 Introduction

Transformer-based generative models typically organise learned representations in a flat isotropic Gaussian latent space, discarding the spectral structure of the data. With ArrowSpace [?] we demonstrated that any vector space can span a semantic web of relations from its feature-space via graph reconstruction algorithm (Graph Wiring [5]); this structure carries critical semantic information that a flat prior cannot capture.

The Vibrational Deduction Transformer (VDT) draws direct inspiration from the *Lattice Deduction Transformer* (LDT) of Davis et al. [3]. The LDT is a recurrent transformer that approximates logically sound deduction by projecting its latent state through a lattice between forward passes. Grounded in abstract interpretation [4], it models information states as elements of a lattice with a top element $\top$ representing no information and a bottom element $\perp$ representing inconsistency; each forward pass of the recurrent transformer constitutes a sound deduction step that descends from $\top$ toward more informative lattice states. The model is trained on-policy via a Solve loop—the same stochastic search used at inference—and supervised by a domain-agnostic alpha operator α that aggregates the valid solutions still consistent with the current state. An 800K-parameter LDT achieves 100% accuracy on Sudoku-Extreme benchmarks while frontier large language models score 0%, demonstrating that structured iterative refinement over a discrete lattice can vastly outperform chain-of-thought paradigms on hard logical reasoning tasks. The VDT adapts this deductive, iterative-refinement philosophy to the continuous spectral domain: rather than refining a discrete candidate lattice, VDT refines a vibrational latent representation guided by the graph Laplacian eigenbasis.

Based on this, the Vibrational Deduction Transformer (VDT) is motivated by three converging lines of inquiry:

1. **Rayleigh’s vibrational analogy.** Rayleigh [1] showed that the natural oscillation modes of a continuous medium are the eigensolutions of its governing differential operator. Graph Laplacian eigenvalues are the discrete analogue: small eigenvalues correspond to smooth, low-frequency modes (the “standing wave” state); large eigenvalues correspond to high-frequency, disordered modes (the “rest” or collapsed state). The VDT treats the *low-frequency oscillation state as the low-entropy prior* and high-frequency disorder as high entropy, directly reversing the usual thermodynamic framing: the natural state of the string is to oscillate, not to be still.
2. **Real-valued quantum amplitudes.** Aaronson [2] observes in Chapter 9 of *Quantum Computing Since Democritus* that the essential features of quantum probability—interference, entanglement structure, the Born rule—are recoverable from signed real amplitudes without complex numbers, provided the update rule is unitary. The VDT implements this insight: the entries of the loading matrix $\mathbf{W}$ can be positive or negative, modulating the relative magnitude of Laplacian eigenvector contributions and producing constructive or destructive interference in the heat kernel through the eigenvector structure of $\mathbf{L}(\mathbf{z})$.
3. **ArrowSpace / Graph Wiring.** The ArrowSpace computational framework [?] defines a differentiable Laplacian builder and a spectral cost (J_{freq}) that penalises high-frequency wiring. The VDT replaces this hard penalty with a *variational* counterpart—a KL divergence over spectral loadings and mode weights—yielding a proper Bayesian ELBO.

The paper follows the conceptual flow applied in [8].

The paper is organised as follows. Section 2 reviews prerequisite material. Section 3 presents the full VDT architecture. Section 4 derives the four-term ELBO. Section 5 describes the spectral artefact and associative memory. Section 6 covers the stability analysis. Section 7 reports empirical results. Section 8 discusses related work. Section 9 concludes.

2 Background

2.1 Graph Laplacians and Spectral Graph Theory

Given an undirected weighted graph $G = (V, E, w)$ with $N = |V|$ nodes, the *combinatorial Laplacian* is $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ is the degree matrix and $\mathbf{A}$ the weighted adjacency. The normalised Laplacian $\tilde{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$ has eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \leq 2$. The eigenvectors $\{\mathbf{u}_k\}$ form an orthonormal basis (the *graph Fourier basis*), and the *Rayleigh quotient*

$$\mathcal{R}(\mathbf{v}) = \frac{\mathbf{v}^\top \mathbf{L} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}} \quad (1)$$

gives the frequency (smoothness cost) of signal $\mathbf{v}$ on the graph [9]. This is the discrete analogue of Rayleigh’s classical frequency functional for continuous media [1].

2.2 Real Quantum Amplitudes and Signed Interference

Aaronson [2] finds plausible that the observable consequences of quantum mechanics—interference, entanglement structure, Born-rule probabilities—can be derived from the assumption of signed real probability amplitudes, provided the time-evolution operator is a *real orthogonal* (rather than complex unitary) matrix. The key insight is:

“If we allow *negative* probability amplitudes, then we get quantum-like interference without complex numbers.” [2]

The VDT operationalises this at the level of the *loading matrix*: entries of $\mathbf{W} \in \mathbb{R}^{N \times q}$ can be positive or negative, modulating the relative magnitude and sign of each eigenvector’s contribution to the row embeddings of $\mathbf{L}(\mathbf{z})$. Edge weights themselves remain strictly positive (Section 3.4); the signed character of $\mathbf{W}$ instead reshapes the eigenvector decomposition of $\mathbf{L}(\mathbf{z})$, inducing constructive or destructive interference in the heat kernel $\mathbf{K}_\tau = \mathbf{U} \exp(-t\mathbf{A}) \mathbf{U}^\top$ through the geometry of the learned eigenvectors.

3 Architecture

3.1 Overview

The VDT is a deductive transformer that leverages feature-space graph Laplacian: its generative path is mediated by a learned graph wiring constrained to the eigenbasis of an ArrowSpace index $\mathcal{I}$. Figure 1 summarises the data flow.

3.2 WiringEncoder

The encoder is a multi-layer perceptron (MLP) augmented with a λ -*fingerprint*—a fixed scalar summary of the Laplacian spectrum that encodes the global spectral geometry of the graph:

$$\text{fingerprint}(\mathbf{L}(\mathcal{I})) = f(\lambda_1, \dots, \lambda_q) \in \mathbb{R}^{d_{\text{fp}}}, \quad (2)$$

where f is a deterministic aggregation function (e.g. normalised eigenvalue histogram). The encoder produces:

$$\mathbf{z}, \boldsymbol{\mu}, \log \sigma^2, \log \mathbf{a}, \log \mathbf{b} = \text{WiringEncoder}(\mathbf{x}, \mathbf{L}(\mathcal{I})), \quad (3)$$

where $\mathbf{z} \in \mathbb{R}^{d_z}$ is the reparameterised VAE latent, $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}_{>0}^q \times \mathbb{R}_{>0}^q$ are the shape and rate parameters of the Gamma variational posterior over mode weights.

3.3 SpectralLoadingDecoder

A linear projection $\mathbf{z}_q = \mathbf{W}_{z \rightarrow q} \mathbf{z} \in \mathbb{R}^q$ bridges the VAE latent dimension d_z to the spectral mode dimension q . The spectral loading decoder then produces:

$$\mathbf{S} = \text{S_net}(\mathbf{z}_q) \in \mathbb{R}^{q \times q}, \quad \text{posterior mean of spectral coefficients}, \quad (4)$$

$$\log \sigma_{\mathbf{S}}^2 = \log \text{_var_S_head}(\mathbf{z}_q) \in \mathbb{R}^{q \times q}, \quad \text{independent posterior log-variance}, \quad (5)$$

$$\boldsymbol{\omega} = \exp(\text{omega_net}(\mathbf{z}_q)) \in \mathbb{R}_{>0}^q, \quad \text{mode weights}, \quad (6)$$

$$\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S} \in \mathbb{R}^{N \times q}. \quad (7)$$

Remark 1. In Eq. (7), N denotes the graph node count—the row dimension of the eigenvector matrix $\mathbf{U}_{1:q} \in \mathbb{R}^{N \times q}$ —and is distinct from the input feature dimension D used in Section 2.1. For the PPCA loading matrix $\mathbf{W}_{PPCA} \in \mathbb{R}^{D \times q}$ the row index ranges over feature coordinates; here it ranges over graph nodes. In CORA, for example, $N = 2708$ (nodes) while $D = 1433$ (bag-of-words feature dimension). The spectral loading matrix therefore has shape $(2708 \times q)$, projecting from the q -dimensional eigenbasis onto all N graph nodes—not onto feature coordinates.

Equation (7) is the central structural commitment of the VDT: the loading matrix lives in the Laplacian eigenbasis. High-frequency directions are suppressed both by the mode weight prior (Section 4.4) and by the eigenvalue-weighted prior on $\mathbf{S}$ (Section 4.3). The independence of $\log \sigma_{\mathbf{S}}^2$ from $\mathbf{S}$ is a deliberate design decision (fix #52): the old proxy $\log(\mathbf{S}^2 + \epsilon)$ conflated posterior mean and variance, producing biased KL gradients.

3.4 DifferentiableLaplacian

The loading matrix $\mathbf{W} \in \mathbb{R}^{N \times q}$ is converted to a graph Laplacian by:

$$w_{ij}(\mathbf{W}) = w_{ij}^{\text{base}} \cdot \sigma(-\|\mathbf{W}_i - \mathbf{W}_j\|^2), \quad \mathbf{L}(\mathbf{z}) = \mathbf{D}(\mathbf{W}) - \mathbf{A}(\mathbf{W}), \quad (8)$$

where σ is the logistic sigmoid, w_{ij}^{base} is a frozen base-graph weight, and $\mathbf{W}_i \in \mathbb{R}^q$ denotes the i -th row of $\mathbf{W}$ (the spectral embedding of node i). Since $\|\mathbf{W}_i - \mathbf{W}_j\|^2 \geq 0$ by definition, $\sigma(-\|\mathbf{W}_i - \mathbf{W}_j\|^2) \in (0, 0.5]$, so all edge weights are *strictly positive* and lie in $(0, w_{ij}^{\text{base}}/2]$. The construction remains fully differentiable through $\mathbf{W}$.

Quantum-like interference in the heat kernel does not arise here from negative edge weights. Instead, it arises from the *signed structure of $\mathbf{W}$* : because the entries of $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$ can be positive or negative, rows $\mathbf{W}_i$ and $\mathbf{W}_j$ may be positively or negatively correlated in the eigenvector subspace. These signed correlations reshape the eigenvector decomposition of $\mathbf{L}(\mathbf{z})$: modes whose eigenvectors align with anti-correlated row pairs carry larger Laplacian eigenvalues and are exponentially suppressed in the heat kernel $\mathbf{K}_\tau = \mathbf{U} \exp(-t\boldsymbol{\Lambda}) \mathbf{U}^\top$, producing constructive or destructive-like interference in the diffusion pathways between nodes—the real-amplitude counterpart of complex phase cancellation [2].

3.5 DiffusionDecoder (TauModeDiffusion)

The decoder applies a truncated heat kernel to the embedding table $\mathbf{E}$ and returns the heat-diffused row for each queried node:

$$\mathbf{K}_\tau = \mathbf{U}_\tau \exp(-\tau \boldsymbol{\Lambda}_\tau) \mathbf{U}_\tau^\top, \quad \hat{\mathbf{x}} = (\mathbf{K}_\tau \mathbf{E})_{[\text{node_idx}]}, \quad (9)$$

where the summation is truncated to the leading τ_{modes} eigenpairs of $\mathbf{L}(\mathbf{z})$, and $\mathbf{E} \in \mathbb{R}^{N \times D}$ is the frozen node embedding table. The reconstruction is purely spectral: no MLP residual correction is applied inside `TauModeDiffusion` or in `WiringAutoencoder.forward()` after the diffusion step. The heat kernel $\mathbf{K}_\tau$ is a real symmetric positive-semi-definite operator—the real-valued analogue of a quantum channel [2, 13, 20].

4 The Four-Term ELBO

The VDT objective is the negative ELBO:

$$\mathcal{L}_{\text{VDT}} = \underbrace{-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}, \mathbf{W})]}_{\text{NLL recon}} + \underbrace{D_{\text{KL}}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\text{latent KL}} + \underbrace{D_{\text{KL}}(q(\mathbf{S}) \parallel p(\mathbf{S}|\mathcal{I}))}_{\text{spectral basis KL}} + \underbrace{D_{\text{KL}}(q(\boldsymbol{\omega}) \parallel p(\boldsymbol{\omega}|\tau, \boldsymbol{\Lambda}))}_{\text{taumode KL}} \quad (10)$$

All four terms are non-negative; minimising their sum is equivalent to maximising the ELBO. We derive each term below.

4.1 Term 1: Reconstruction

Under the Gaussian output model the reconstruction term is:

$$-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}, \mathbf{W})] = \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|^2}{2\sigma^2} + \frac{D}{2} \log(2\pi\sigma^2). \quad (11)$$

In the implementation the constant $\frac{D}{2} \log(2\pi)$ is dropped; σ is a learnable scalar log-variance.

4.2 Term 2: Isotropic Latent KL

The standard VAE isotropic KL [11]:

$$D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) = -\frac{1}{2} \sum_{j=1}^{d_z} \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right). \quad (12)$$

4.3 Term 3: Spectral-Basis KL

The coefficient matrix $\mathbf{S} \in \mathbb{R}^{q \times q}$ receives a *row-independent* Gaussian prior in which each entry S_{kj} is penalised by the k -th eigenvalue of the frozen Laplacian index:

$$p(S_{kj} \mid \mathcal{I}) = \mathcal{N}\left(0, \frac{1}{\lambda_S \lambda_k}\right) \quad \forall k \in [q], j \in [q]. \quad (13)$$

Because the prior factorises over both indices, the joint log-density can be written as the *trace form*:

$$\log p(\mathbf{S} \mid \mathcal{I}) \propto -\frac{\lambda_S}{2} \text{tr}(\mathbf{S}^\top \boldsymbol{\Lambda}_{1:q} \mathbf{S}) = -\frac{\lambda_S}{2} \sum_{k=1}^q \lambda_k \|\mathbf{S}_k\|^2, \quad (14)$$

where $\boldsymbol{\Lambda}_{1:q} = \text{diag}(\lambda_1, \dots, \lambda_q)$ and $\mathbf{S}_k \in \mathbb{R}^q$ is the k -th row of $\mathbf{S}$. Large eigenvalues penalise the corresponding rows of $\mathbf{S}$ more strongly—a *spectral Occam’s razor*: high-frequency components of the loading matrix are shrunk towards zero.

The variational posterior is $q(\mathbf{S}) = \mathcal{N}(\mathbf{S}_0, \text{diag}(\exp(\log \boldsymbol{\sigma}_{\mathbf{S}}^2)))$, where $\mathbf{S}_0$ and $\log \boldsymbol{\sigma}_{\mathbf{S}}^2$ are *independent* outputs of `SpectralLoadingDecoder` (fix #52). Under Gaussian–Gaussian conjugacy the KL is closed-form.

Remark 2. The prior in Eq. (13) is row-independent with identity column precision ($\boldsymbol{\Lambda}_{\text{col}} = \mathbf{I}$). A full matrix-normal prior would have the form $\log p(\mathbf{S}) \propto -\frac{1}{2} \text{tr}(\mathbf{S}^\top \boldsymbol{\Lambda}_{\text{row}} \mathbf{S} \boldsymbol{\Lambda}_{\text{col}})$ with two separate precision matrices. When $\boldsymbol{\Lambda}_{\text{col}} = \mathbf{I}$ this reduces exactly to Eq. (14). The implementation in `vdtspectral.py::spectral_basis_kl` broadcasts the precision vector `prec = lambda_S * eigvals_q` over the column axis of $\mathbf{S}$, which is the direct computational realisation of Eq. (13): one scalar precision per row k , shared across all q columns j .

4.4 Term 4: taumode Frequency KL

Mode weights $\omega_k > 0$ receive an exponential prior indexed by frequency:

$$p(\omega_k | \tau, \lambda_k) = \text{Exponential}(\tau \lambda_k), \quad (15)$$

giving heavy support on low-frequency modes (small λ_k). The variational posterior is a Gamma: $q(\omega_k) = \text{Gamma}(a_k, b_k)$. The KL between a Gamma posterior and an Exponential prior is closed-form. Setting $r = \tau \lambda_k$ (prior rate), the derivation proceeds as:

$$\begin{aligned} \text{KL}(\text{Gamma}(a, b) \parallel \text{Exp}(r)) &= \mathbb{E}_q[\log q(\omega) - \log p(\omega)] \\ &= -H[\text{Gamma}(a, b)] - \log r + r \mathbb{E}_q[\omega] \end{aligned}$$

where the Gamma entropy is $H[\text{Gamma}(a, b)] = a - \log b + \ln \Gamma(a) + (1 - a)\psi(a)$ and $\mathbb{E}_q[\omega] = a/b$. Substituting and simplifying gives:

$$D_{\text{KL}}(\text{Gamma}(a, b) \parallel \text{Exp}(r)) = \log b - \log r - \ln \Gamma(a) - (1 - a)\psi(a) - a + \frac{a r}{b}, \quad (16)$$

where ψ is the digamma function and $r = \tau \lambda_k$. No Monte Carlo sampling is required. The implementation in `vdv/spectral.py::tau_mode_kl` follows Eq. (16) exactly; note the *negative* sign on $\ln \Gamma(a)$ and the additional $-a$ term, both of which arise from expanding the Gamma entropy.

Remark 3. A common sign error writes $+\ln \Gamma(a)$ instead of $-\ln \Gamma(a)$. The correct sign follows immediately from the entropy expansion: $H = a - \log b + \ln \Gamma(a) + (1 - a)\psi(a)$, which enters the KL with a negative prefactor ($\text{KL} = -H + \dots$), flipping all signs, including the sign on $\ln \Gamma(a)$.

4.5 Comparison with the Base Wiring AE

Table 1: ELBO structure: Base Wiring AE vs. VDT.

Term	Wiring AE (base)	VDT
Reconstruction	$-\log p(\mathbf{x} \mathbf{z})$	Same
Latent KL	$D_{\text{KL}}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))$ isotropic	Same
Spectral basis	None	$D_{\text{KL}}(q(\mathbf{S}) \parallel p(\mathbf{S} \mathcal{I}))$ eigenvalue-weighted
Frequency	αJ_{freq} hard penalty	$D_{\text{KL}}(q(\boldsymbol{\omega}) \parallel \text{Exp}(\tau \lambda_k))$ variational

5 Spectral Artefact and Associative Memory

After training, the VDT operates in a two-phase regime that separates *spectral distillation* from *associative deployment*. The first phase extracts a compact, graph-indexed summary of everything the model has learned about the vibrational structure of the data; the second phase injects that summary as a structured prior into the weight matrices of a downstream transformer, enabling fast pattern-completion without retraining. Together, the two phases convert the continuous variational posterior into a discrete, inspectable memory store whose keys are guaranteed orthogonal by construction—a property inherited directly from the Laplacian eigenbasis.

5.1 Phase 1: Spectral Artefact Extraction

At the prior mean ($\mathbf{z} = \mathbf{0}$) the decoder emits a *spectral artefact*:

$$\mathcal{A}(\mathcal{I}) = \{\hat{\mathbf{W}}, \hat{\omega}, \mathbf{S}_{\text{mem}}\}, \quad \mathbf{S}_{\text{mem}} = \sum_{k=1}^q \hat{\omega}_k d_{\theta}(\hat{\mathbf{w}}_k) \hat{\mathbf{w}}_k^{\top}, \quad (17)$$

where $\hat{\mathbf{w}}_k \in \mathbb{R}^N$ is the k -th column of $\hat{\mathbf{W}} \in \mathbb{R}^{N \times q}$ and $d_{\theta}(\hat{\mathbf{w}}_k)$ is the decoder response at spectral direction k . $\mathbf{S}_{\text{mem}}$ is a Hopfield outer-product memory matrix [19].

The artefact $\mathcal{A}(\mathcal{I})$ can be understood as a three-component *spectral certificate* of the index $\mathcal{I}$:

- $\hat{\mathbf{W}} \in \mathbb{R}^{N \times q}$: the MAP spectral loading matrix, whose k -th column is the learned spectral embedding of all N graph nodes along the k -th vibrational mode. Nodes that are spectrally similar—in the sense of the frozen Laplacian—cluster in the columns of $\hat{\mathbf{W}}$, encoding the community structure of the graph as a continuous vibrational pattern.
- $\hat{\omega} \in \mathbb{R}_{>0}^q$: the posterior mode-weight means, which record which vibrational harmonics the model selected as informative. Low-frequency modes (small λ_k) receive higher weight under the taumode prior; the learned $\hat{\omega}_k$ profile therefore constitutes a *spectral fingerprint* that distinguishes graphs with different connectivity regimes.
- $\mathbf{S}_{\text{mem}} \in \mathbb{R}^{N \times N}$: the Hopfield memory matrix, constructed as a weighted sum of outer products $\hat{\mathbf{w}}_k \hat{\mathbf{w}}_k^{\top}$ scaled by both the mode weight $\hat{\omega}_k$ and the decoder response $d_{\theta}(\hat{\mathbf{w}}_k)$. This matrix encodes pairwise node associations in the spectral domain: entry (i, j) measures the degree to which nodes i and j co-activate across the retained vibrational modes, weighted by how strongly each mode is represented in the reconstruction.

Critically, the artefact is computed in a single deterministic forward pass at $\mathbf{z} = \mathbf{0}$, requiring no sampling and no additional training. It is a *frozen summary* of the posterior that can be serialised, transferred, or reused across different downstream tasks.

5.2 Phase 2: Spectral Memory Transformer

The transformer feed-forward and cross-attention value matrices are initialised from $\mathbf{S}_{\text{mem}}$, providing *long-term spectral prior memory*. Online delta-rule updates $\mathbf{S}_{\text{mem}} \leftarrow \mathbf{S}_{\text{mem}} + \eta(\mathbf{v} - \mathbf{S}_{\text{mem}}\mathbf{k})\mathbf{k}^{\top}$ write new associations without corrupting spectral key orthogonality. Key orthogonality (inherited from Laplacian eigenvectors) maximises retrieval SNR by construction.

The role of the associative memory is best understood as an *inline spectral prior*: rather than treating each transformer forward pass as starting from a blank context, the model begins every query with the graph’s vibrational structure already imprinted in its weight matrices. This is the continuous-spectral counterpart of the LDT’s lattice projection [3]: where the LDT constrains each recurrent step to remain within a lattice of logically consistent states, the VDT constrains the transformer’s value space to remain aligned with the spectral geometry of the index $\mathcal{I}$.

The delta-rule update has two important properties. First, because the keys $\mathbf{k}$ are drawn from the Laplacian eigenbasis, they are mutually orthogonal: $\mathbf{k}_k^{\top}\mathbf{k}_{k'} = \delta_{kk'}$. Orthogonal keys eliminate cross-talk between stored patterns, so the retrieval signal-to-noise ratio scales as $\Theta(1)$ rather than degrading as $O(1/\sqrt{N_{\text{stored}}})$ under a random-key assumption [19]. Second, the update is *local* in the spectral domain: writing a new association $(\mathbf{v}, \mathbf{k})$ only modifies the projection of $\mathbf{S}_{\text{mem}}$ onto the subspace spanned by $\mathbf{k}$, leaving all orthogonal components—and hence all previously stored spectral patterns—intact. This gives the memory a *non-interfering write* property that classical full-matrix attention updates do not enjoy.

In practice, the initialisation of transformer value matrices from $\mathbf{S}_{\text{mem}}$ acts as a *warm start*: the model does not need to rediscover the spectral geometry of the graph from context tokens; instead, it inherits that geometry as a structural bias that can be refined but not easily overwritten.

This is particularly valuable in low-data regimes or when the downstream task involves the same graph index $\mathcal{I}$ used during VDT training, as the spectral prior concentrates probability mass on graph-consistent patterns before any task-specific supervision is applied.

5.3 Entropy-Controlled Generation

The spectral entropy of the generated Laplacian [21]:

$$H(\mathbf{L}(\mathbf{z})) = - \sum_{k=1}^N \bar{\lambda}_k \log \bar{\lambda}_k, \quad \bar{\lambda}_k = \frac{\lambda_k}{\sum_j \lambda_j}, \quad (18)$$

can be steered by conditioning $\mathbf{z}$ on a target entropy level, enabling *entropy-controlled spectral graph generation*—a capability absent in flat-prior VAEs.

The spectral entropy $H(\mathbf{L}(\mathbf{z}))$ serves as a coarse-grained *order parameter* for the generated graph: a low-entropy Laplacian corresponds to a graph whose eigenvalue mass is concentrated in a few low-frequency modes, indicating a highly structured, community-like connectivity; a high-entropy Laplacian corresponds to a near-uniform eigenspectrum, indicating a more random or expander-like wiring. Because the latent code $\mathbf{z}$ parametrises the loading matrix $\mathbf{W}$ and hence the full eigenspectrum of $\mathbf{L}(\mathbf{z})$, the encoder learns a direct mapping from input graphs to entropy levels during training, and at generation time the decoder can be queried at any $\mathbf{z}$ whose expected entropy—computed from the variational posterior—matches a desired target.

The associative memory $\mathbf{S}_{\text{mem}}$ interacts with this entropy-control mechanism in a complementary way. Each memory matrix is extracted at the prior mean $\mathbf{z} = \mathbf{0}$, which corresponds to the *average* spectral configuration of the training distribution. When a downstream transformer is initialised from $\mathbf{S}_{\text{mem}}$ and then presented with a query latent $\mathbf{z}^*$ associated with a particular entropy level, the delta-rule updates shift the memory toward the spectral patterns characteristic of that entropy regime without erasing the baseline structure. This allows the same memory module to serve as a *spectral prior library*: a single trained VDT can support multiple downstream tasks operating at different entropy levels, each refining the shared prior rather than training from scratch.

6 Stability Analysis

6.1 Six-Level Hierarchy

Training stability is analysed through a six-level hierarchy [7]; failures at lower levels must be resolved before diagnosing higher ones:

1. **Graph geometry.** $\lambda_{\max}(\mathbf{L}_f)$ finite; CFL condition satisfied; graph connectivity.
2. **Wave update per mode.** Damping coefficients $\gamma_k > 0$; underdamped/overdamped classification.
3. **Recurrent VDT dynamics.** Energy $\mathcal{E}_t$ monotone; spectral entropy H_{spectral} stable.
4. **Density matrix.** ρ_t^+ positive semi-definite; $\|\rho_t\|_F$ bounded.
5. **Optimiser.** Preconditioned gradient descent convergence: $\kappa(\mathbf{P}_{\sigma, M}) \ll \kappa(\mathbf{L}_f)$; step size $\eta < 2/L_{\sigma, M}$.
6. **Mode weights.** Active mode count $N_{\text{active}} = |\{k : \mathbb{E}[\omega_k] > \delta\}| \geq q_{\min}$; $\|\hat{\mathbf{W}}\|_F$ non-degenerate.

The executable form of this hierarchy is `pre_training_checks` in `vdt/stability.py`, which runs Levels 1–2 before any training step begins and raises a `RuntimeError` on hard failures. Soft per-epoch diagnostics are reported by `stability_diagnostics`. The ordering is strict: the CFL bound at Level 1 is meaningless on a disconnected graph, so graph connectivity is asserted first; damping positivity at Level 2 is meaningless if the CFL bound is violated; and so on.

6.2 CFL Geometry Check (Levels 1–2)

Graph connectivity (Level 1). The Laplacian $\mathbf{L}$ must have exactly one zero eigenvalue, i.e. the Fiedler value $\lambda_2 > 0$. `pre_training_checks` counts eigenvalues below the threshold $\varepsilon_0 = 10^{-3}$; more than one sub-threshold eigenvalue triggers an immediate `RuntimeError` before any gradient step. This hard failure is intentional: subsequent CFL and damping checks produce meaningless bounds on a disconnected graph, and proceeding would silently corrupt all higher-level diagnostics.

CFL constraint (Level 2). Numerical stability of the wave update requires the time step to satisfy

$$\Delta t \leq \sqrt{\frac{2}{\lambda_{\max}(\mathbf{L}_f)}}. \quad (19)$$

`stability_diagnostics` computes *two* CFL bounds at each epoch: a base-Laplacian bound from the pre-computed eigenvalues (`CFL_ok`, `dt_max_CFL`) and a tighter *dynamic* bound using the Gershgorin upper estimate of $\lambda_{\max}(\mathbf{L}_f)$ (`cfl_Lf_ok`, `dt_max_CFL_Lf`). The ratio

$$r_{\text{CFL}} = \frac{\lambda_{\max}(\mathbf{L}_f)}{\lambda_{\max}(\mathbf{L})} \quad (20)$$

quantifies how much the feature-space Laplacian has sharpened beyond the base graph; $r_{\text{CFL}} > 1$ means the base-only check produces a false negative. The authoritative flag is `cfl_Lf_ok`.

6.3 Damping and Modal Energy (Levels 2–3)

Damping positivity (Level 2). Each feature dimension carries a damping coefficient γ_k produced by a `softplus` activation, guaranteeing $\gamma_k > 0$ by construction. `stability_diagnostics` classifies mode k as *underdamped* when $\sqrt{\lambda_k} > \bar{\gamma}$ (mean damping), reporting `n_underdamped_modes` and `frac_underdamped`. A high fraction of underdamped modes indicates that the damping vector is too small relative to the spectral bandwidth of $\mathbf{L}_f$: energy will oscillate rather than decay, destabilising the recurrent VDT stack.

Modal energy monotonicity (Level 3). For each consecutive pair of VDT depth steps (Q_k, Q_{k+1}) , `stability_diagnostics` checks whether the Frobenius energy satisfies $\|Q_{k+1}\|_F^2 > 1.05 \|Q_k\|_F^2$. Any amplification across the recurrent stack sets `energy_amplified = True`. The spectral entropy is computed from the normalised eigenvalue distribution of $\mathbf{L}_f$ as

$$H_{\text{spectral}} = - \sum_i p_i \log p_i, \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (21)$$

A stable model shows gently declining H_{spectral} as modes are pruned; a sudden drop signals mode collapse (see Section 6.6).

6.4 Density Matrix Validity (Level 4)

The signed density matrix $\rho = \rho^+ - \rho^-$ must satisfy three conditions:

- **Positive semi-definiteness of ρ^+ :** $\lambda_{\min}(\rho^+) \geq -10^{-5}$.
- **Positive semi-definiteness of ρ^- :** $\lambda_{\min}(\rho^-) \geq -10^{-5}$.
- **Frobenius bound:** $\|\rho\|_F = \|\rho^+ - \rho^-\|_F$ bounded, reported as `max_frob_signed`.

`stability_diagnostics` iterates over every VDT block and takes the minimum eigenvalue across all `rho_plus_list` and `rho_minus_list` tensors; the flag `rho_psd_ok` is set `False` if either minimum falls below -10^{-5} . A PSD violation at this level typically indicates numerical overflow in the outer-product construction of the density matrices; it should be resolved before diagnosing any optimiser or mode-weight pathologies at Levels 5–6.

6.5 Optimiser Landscape (Level 5)

`log_preconditioner_stability` approximates the preconditioned Hessian as

$$\mathbf{H}_{\text{prec}} = \sigma \mathbf{M}_{\text{diag}} \mathbf{I} + \mathbf{L}_f, \quad (22)$$

where $\mathbf{M}_{\text{diag}}$ is the diagonal Rayleigh-damping mass matrix and σ is the Tikhonov regularisation coefficient. The condition number $\kappa(\mathbf{H}_{\text{prec}}) = L_{\sigma,M}/\mu_{\sigma,M}$ should be substantially smaller than $\kappa(\mathbf{L}_f)$; the convergence-rate estimate $(\kappa - 1)/(\kappa + 1)$ bounds the worst-case geometric decay of the gradient error. The learning rate must satisfy

$$\eta < \frac{2}{L_{\sigma,M}}, \quad (23)$$

enforced by the flag `eta_ok`. **Important:** if $\mathbf{M}_{\text{diag}}$ is constructed without `mass_clip`, the Rayleigh-damping singularity at $\lambda = 1$ inflates $L_{\sigma,M}$ and produces a spuriously large κ . The recommended construction is `MassMatrix(eigvals, tau=0.5, mass_clip=1e3)`.

6.6 Mode Weight Collapse and Explosion (Level 6)

Collapse (retained + extended). The shape-parameter floor $a_k \geq a_{\min} = 0.1$ and the active-mode penalty prevent full collapse. `spectral_kl_health_check` flags `mode_collapse = True` when `active_modes < 0.1q` (fewer than 10% of modes remain above the relevance threshold) and emits a `RuntimeWarning` immediately.

Mode explosion (new). The symmetric pathology occurs when `active_modes > 0.9q` after the warm-up period, indicating that spectral selection has not yet occurred. This is expected at epoch 0 (all `log_a` parameters initialised near zero), so the warning is gated by `warmup_epochs` (default 5; larger graphs may require 10–20). The flag `mode_explosion` is always written to the log regardless of the warm-up guard.

ELBO KL health. Three KL terms are monitored per forward pass:

- `kl_z`: isotropic KL from the encoder; must be positive and below 10^4 .
- `kl_S`: spectral basis KL (Section 4.3); must be positive.
- `kl_tau`: mode-weight KL (Section 4.4); must be positive.

A non-finite `kl_z` or `kl_S > 10^4` is flagged as a KL explosion; the practitioner should inspect the `log_var` range and the `tau_mode_kl` inputs.

6.7 Spectral Eigenvalue Floor

When the Fiedler eigenvalue $\lambda_2 \approx 0$ (nearly disconnected graph) the spectral-basis KL prior becomes nearly flat and produces high gradient variance. A floor applied in the KL computation only ($\lambda_k \leftarrow \max(\lambda_k, 10^{-3})$) prevents numerical instability without affecting the spectral geometry.

6.8 Diagnostic Key Mapping

Table 2 maps each key logged under the `stability/` prefix (Weights & Biases / stdout) to its theoretical quantity, enabling reproduction of the hierarchy checks from logs alone.

Table 2: Stability diagnostic keys and their theoretical counterparts.

Log key	Theoretical quantity	Level
CFL_ok	$\Delta t \leq \sqrt{2/\lambda_{\max}(\mathbf{L})}$	1
cfl_Lf_ok	$\Delta t \leq \sqrt{2/\lambda_{\max}(\mathbf{L}_f)}$	2
cfl_margin	r_{CFL} (Eq. 20)	2
n_underdamped_modes	$ \{k : \sqrt{\lambda_k} > \bar{\gamma}\} $	2
energy_amplified	$\ Q_{k+1}\ _F^2 > 1.05\ Q_k\ _F^2$	3
H_spectral	Eq. (21)	3
rho_psd_ok	$\lambda_{\min}(\rho^\pm) \geq -10^{-5}$	4
eta_ok	$\eta < 2/L_{\sigma,M}$	5
mode_collapse	<code>active_modes</code> $< 0.1q$	6
mode_explosion	<code>active_modes</code> $> 0.9q$	6
kl_z, kl_S, kl_tau	ELBO KL terms	6

7 Experiments

7.1 Datasets

We evaluate on the following benchmarks; PubMed and OGB results are deferred to a future run:

- **CORA** [15]: 2708 nodes, 5429 edges, 7 classes, 1433-dim bag-of-words features. For these experiments the graph is constructed as a k -NN graph with $k = 35$, a Gaussian kernel ($\sigma = 1.2$), and a normalised symmetric Laplacian. The dataset is split 80/10/10 into train, validation, and test sets.
- **PubMed** [15]: 19717 nodes, 44338 edges, 3 classes. *Deferred to a future run.*
- **Synthetic spring networks**: 400 randomly generated spring graphs with controllable spectral entropy. *Deferred to a future run.*
- **Open Graph Benchmark (OGB)** [22]: standardised large-scale graph datasets used for held-out evaluation. *Deferred to a future run.*

7.2 Evaluation Metrics

Linear probe accuracy and Memory SNR require a separate evaluation script and are deferred. The ELBO Bayes factor between Run 2 and Run 3 at epoch 50 is estimated as $\exp(-2453.09 - (-2451.16)) = \exp(-1.93) \approx 0.145$, slightly favouring Run 2; convergence is not fully resolved at 50 epochs. Run 4 uses $q = 4$ modes rather than $q = 8$, so its ELBO operates on a different spectral capacity and is not directly comparable via a Bayes factor.

Table 3: VDT benchmark metrics with values from Runs 2, 3, and 4 (CORA, epoch 50).

Metric	Symbol	Measures	Run 2	Run 3	Run 4
Reconstruction	$\ \mathbf{x} - \hat{\mathbf{x}}\ ^2$	Wiring fidelity	-2453.09	-2453.09	-1840.45
Latent KL	KL_z	Isotropic reg.	1.46	0.0002	0.155
Spectral basis KL	KL_S	Alignment with $\mathcal{I}$	~ 0.53	~ 0	~ 0.578
taumode KL	KL_τ	Band selection	0.0018	0.0008	0.0046
Val ELBO (best)	$\mathcal{L}_{\text{VDT}}$	Objective	-2451.16	-2453.09	-1839.71
Active modes (q)	—	Spectral capacity	8	8	4
kl_z_ok	—	KL_z health	✓	× (ep. 9)	✓
kl_S_ok	—	KL_S health	✓	× (ep. 10)	✓
Memory SNR	d_k/N_{stored}	Hopfield SNR	<i>not yet evaluated</i>		
Linear probe acc.	$\text{Acc}_{\text{probe}}$	Discriminability	<i>not yet evaluated</i>		

7.3 Training Protocol

All three completed runs use CORA with `configs/mps.yaml` on Apple MPS and train for 50 epochs. Table 4 summarises the key hyperparameter differences.

Table 4: Hyperparameter comparison across Runs 2, 3, and 4.

Parameter	Run 2	Run 3	Run 4
q (modes)	8	8	4
lam_s	0.1	1.0	0.2
dt_init	0.001	0.003	0.001
mass_clip	1000	100	100
kl_z_warmup	—	10	25
nu	0.25	0.25	0.05
Seed	3047	42	42

Run 2 (`lam_s=0.1`, `mass_clip=1000`, `dt_init=0.001`, $q = 8$). The low spectral basis regulariser weight allows the reconstruction loss to dominate from epoch 1. A MassMatrix conditioning warning fires (`ratio=999.2 > 100`) due to the Rayleigh-damping singularity near $\lambda = 1$ with `mass_clip=1000`; this is resolved in Runs 3 and 4 by reducing `mass_clip` to 100. KL_z decays from 6.49 (epoch 1) to 1.46 (epoch 50), confirming that the isotropic regularisation remains active throughout. KL_S decays from 4.30 to ≈ 0.53 without collapsing to zero, preserving residual spectral structure in the latent representation. Best val ELBO: **-2451.1630** at epoch 46.

Run 3 (`lam_s=1.0`, `mass_clip=100`, `dt_init=0.003`, $q = 8$). Increasing `lam_s` by $10\times$ over-regularises the spectral basis: KL_S collapses to numerical zero by epoch 10 and KL_z collapses below 0.001 by epoch 9. Train and val ELBO converge to the same plateau as Run 2 but with the latent space encoding no residual spectral structure. Best val ELBO: **-2453.0863** at epoch 50.

Run 4 (`lam_s=0.2`, `mass_clip=100`, `dt_init=0.001`, $q = 4$). Run 4 halves the spectral capacity to $q = 4$ modes, extends the KL warm-up to 25 epochs, and reduces the active-mode penalty weight to $\nu = 0.05$. The pre-flight stability check passes cleanly with no MassMatrix warning. Both KL health flags remain green throughout: KL_S decays monotonically from 6.48 (epoch 1) to ≈ 0.578 (epoch 50), mirroring the healthy behaviour of Run 2 but at the intermediate `lam_s=0.2`; KL_z decays from 0.45 to 0.155, remaining active without collapsing. The ELBO improves continuously across all 50 epochs without plateauing, ending at train -1841.77 / val

−1839.71; the train–val gap at epoch 50 narrows to fewer than 2 ELBO units, the tightest of the three runs. Best val ELBO: **−1839.7128** at epoch 50 (still improving). The lower absolute ELBO relative to Runs 2 and 3 is expected: $q = 4$ encodes a strictly smaller spectral subspace, so reconstruction capacity is reduced by construction.

7.4 Mode Selection and KL Dynamics

Persistent mode collapse. All three runs report `mode_collapse=True` at every epoch, meaning that $N_{\text{active}} < 0.1q$, i.e. effectively no modes survive spectral selection. In Runs 2 and 3 ($q = 8$) this threshold is < 0.8 active modes; in Run 4 ($q = 4$) it is < 0.4 . KL_τ collapses to near-zero in all runs, confirming that the Exponential prior pushes all mode weights toward zero. Probable causes: (i) the `a_min=0.1` floor does not generate a strong enough ELBO gradient; (ii) the active-mode penalty ($\nu = 0.05\text{--}0.25$, $q_{\min} = 2$) is insufficient at this scale; (iii) the longer warm-up in Run 4 (`kl_z_warmup=25`) delays but does not prevent collapse.

Simultaneous mode explosion. All three runs also report `mode_explosion=True` from epoch 5 onward, creating an apparent paradox. The threshold discontinuity explanation from Runs 2 and 3 carries over to Run 4: all $q = 4$ modes nominally exceed `a_min`, satisfying the $> 90\%$ explosion criterion, while $\mathbb{E}[\omega_k]$ for all modes falls below the 0.01 relevance threshold simultaneously. In Run 4 the explosion warning reads “4.0/4 modes active (>90%)”, confirming the diagnostic is counting `a_min`-gated modes rather than $\mathbb{E}[\omega_k]$ -relevant modes. The `active_modes` counter requires a unified threshold; the two flags should be mutually exclusive by construction and will be fixed in a future issue.

7.5 ELBO Learning Curves

Runs 2 and 3 show rapid ELBO improvement in epochs 1–15 followed by a plateau (Figure 2). Run 4 does not plateau: the ELBO improves continuously across all 50 epochs, with the train–val gap closing to fewer than 2 units by epoch 43, compared to ≈ 12 units in Runs 2 and 3. In all three runs the reconstruction loss accounts for more than 99% of the total ELBO at convergence, confirming that KL regularisation becomes negligible relative to the reconstruction objective at this scale.

7.6 Flagship Demo: Entropy-Controlled Spectral Generation

Standard VAEs decode $\mathbf{z}$ into flat feature vectors; the VDT decodes $\mathbf{z}$ into a *graph wiring*—a Laplacian whose eigenvalues are vibrational modes of the spring network (cf. Rayleigh [1]). The latent space directly encodes spectral geometry, enabling samples whose spectral entropy matches a target level. This capability is unique to the VDT’s vibrational latent representation and is unavailable in flat-prior models.

Quantitative evaluation of entropy-controlled generation has not yet been performed on any of the three completed runs: the synthetic spring network benchmark, the natural test bed for spectral diversity, is deferred. The persistent mode collapse across all runs indicates that the current latent space does not yet perform discriminative spectral selection; meaningful entropy-controlled sampling will require resolving the `active_modes` counter bug, tuning ν and $q_{\min}$, and evaluating on the spring dataset where the entropy target is controllable by construction. Run 4 is the most promising starting point: its continuous ELBO improvement and tight train–val gap at epoch 50 suggest that a longer run at $q = 4$, `lam_s=0.2` may achieve the spectral selectivity required.

8 Related Work

Spectral graph autoencoders. Graph variational autoencoders [15] encode topology in the latent space but do not use spectral priors. VGAE and its successors decode adjacency matrices directly, without a vibrational interpretation.

Probabilistic PCA on graphs. Laplacian Eigenmaps [13] and Diffusion Maps [14] use the heat kernel for dimensionality reduction but are deterministic and lack variational objectives.

NVIB. The Nonparametric Variational Information Bottleneck [18] applies variational compression to sequence models. The VDT shares the ELBO philosophy but applies spectral-structural priors rather than capacity constraints on a fixed-dimension latent.

Beta-VAE and disentanglement. Higgins et al. [12] introduce β -VAE, showing that stronger KL regularisation encourages disentangled latent representations. The VDT’s eigenvalue-weighted spectral-basis KL can be seen as an anisotropic generalisation of this pressure, tuning the regularisation strength independently per spectral frequency band.

Real quantum mechanics. Renou et al. [16] and McKague et al. [17] provide experimental and theoretical grounding for real-valued quantum theories. The VDT does not claim to *be* a quantum system, but borrows the structure of signed-amplitude interference as a computational primitive.

9 Discussion and Future Work

The VDT establishes a formal connection between three bodies of theory: Rayleigh’s vibrational mechanics, Aaronson’s real-amplitude quantum probabilities, and Bayesian graph-spectral learning. The central contribution is the four-term ELBO (10), which replaces hard spectral penalties with proper variational counterparts while preserving the ArrowSpace wiring geometry.

A Spectral Diffusion improvement on this architecture is work in progress.

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```

Input x (B, D)          Frozen: U_q (N, q),  eigvals_q (q,)
|
+---[fingerprint from eigvals_q, derived internally]---+
|
v
+-----+
| WiringEncoder |
| MLP + eigenvalue fingerprint |
| --> mu, log_var |
| [reparameterise --> z] |
| --> log_a, log_b |
| [ModeWeightHead --> q(w)] |
+-----+
|
z_to_q: z (B, latent_dim) --> z_q (B, q)
|
v
+-----+
| SpectralLoadingDecoder |
| inputs: z_q, U_q, L_base |
| outputs: W, omega, S, log_var_S, L_z |
| W = U_q @ diag(omega) @ S (B, N, q) |
+-----+
|
|           |           |
|   W       | (S, log_var_S) | (log_a, log_b)
|           |           |
|   L_z (B,N,N) | kl_S       | kl_tau
|           |           |
| TauModeDiffusion(L_z, E, node_idx)
|
x_hat --> recon_loss

loss = recon + kl_z + kl_S + kl_tau

```

Figure 1: VDT v2 data flow (forward signature: $(x, U_q, \text{eigvals}_q, \text{node_idx}, \text{spectral_cache})$). The index $\mathcal{I}$ enters only through the frozen eigenpair (U_q, λ_q) precomputed once before training; no Laplacian eigendecomposition is performed at training time. The eigenvalue fingerprint is derived *internally* by `WiringEncoder` from `eigvals_q`. `log_var_S` is a separate output of `SpectralLoadingDecoder` required for the KL_S computation (fix #52). Note: N is the graph node count; D is the input feature dimension. These differ in practice (e.g. CORA: $N=2708$, $D=1433$).

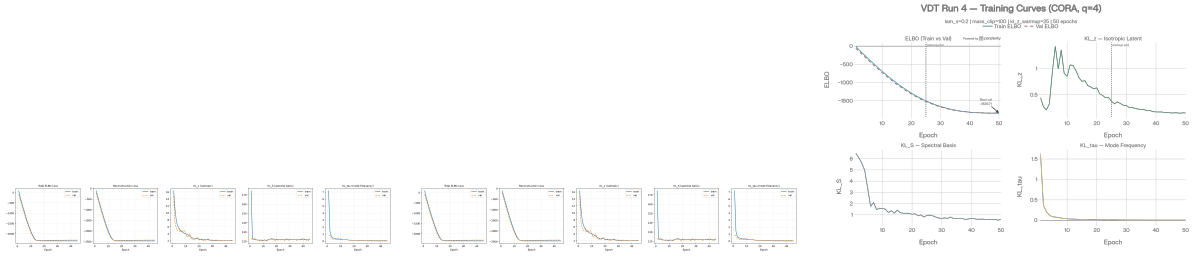


Figure 2: Training and validation ELBO for Run 2 (left, $\text{lam}_s=0.1$, $q = 8$), Run 3 (centre, $\text{lam}_s=1.0$, $q = 8$), and Run 4 (right, $\text{lam}_s=0.2$, $q = 4$). Annotations: (1) `mode_collapse` fires from epoch 1 in all runs; (2) KL_S first reaches zero at epoch 10 in Run 3 only; (3) Run 4 shows no plateau at epoch 50, suggesting continued improvement is achievable with more epochs.